

Edimah SYNESIUS SONGO Mathematical Statistician

 [linkedin.com/in/edimah-synesius](https://www.linkedin.com/in/edimah-synesius)  github.com/edimah
 edimah.synesius-songo@proton.me
 [edimah.github.io](https://github.com/edimah)
 Paris

RESEARCH & PROFESSIONAL EXPERIENCE

Present October 2025	Independent Consultant, APPLIED STATISTICS, AI Implementation > Flagship engagement: interim Data Lead at CRCDC Guadeloupe since February 2026. > Feasibility analysis of AI tools in regulated health care (pharmacy) to facilitate professional workflows. Data Management Communication R SQL eSIS DataQuery Data Visualisation Health Data Regulation
September 2024 November 2023	Doctoral Researcher, IFP ENERGIES NOUVELLES / ONERA, Ile-de-France > Multidisciplinary sensitivity analysis for complex physics systems. > Development of sensitivity indices and explainability metrics under R and Python. > Left the doctorate before completion. R Python Sensitivity Analysis Uncertainty Quantification Multidisciplinary Systems
September 2023 April 2023	Research Intern, EDF R&D, Chatou > M2 Thesis: <i>Advanced methods for model calibration with uncertain inputs.</i> > Designed adaptive, MCMC-based parameter re-adjustment strategies to handle uncertainty. Python OpenTURNS Bayesian Inference Computational Statistics MCMC Algorithms
August 2022 July 2022	Data Analyst Intern, HÔPITAL AVICENNE, Bobigny > Data analysis of patient attendance and trend identification. Tableau Data Analysis Data Visualisation R SQL

RESEARCH & ACADEMIC PROJECTS

DETECTING AGENTIC COLLUSION IN PUBLIC LOGS WITH STATISTICAL METHODS AI Incident Response Sprint, Apart Research × CeSIA, Track 2, solo Detection, dating and characterisation of an LLM-agent swarm from public edit logs alone. Two public sources joined on the same incident: 14,591 reconstructed agent revisions and 32,656 live edits over four wikis, 2003–2026. A ladder of four null models, from a Poisson rate test to a difference-in-differences against an untouched sibling wiki with a placebo-in-time test: the incident weeks exceed all 4,872 reference weeks, the effect is 3.2 log points, and the change point falls on 16 June. A weekly check built from the same features raises no false alarm over the 2,236 wiki-weeks of the preceding decade.  github.com/Edimah/agentic-collusion-detector-in-public-logs Change-Point Detection Difference-in-Differences Placebo Tests Empirical p-values Python	SEPTEMBER 2026
A CALIBRATED TWO-LEVEL DETECTOR FOR SECRET LOYALTIES – APPLICATION TO AN LLM “Secret Loyalties” hackathon, Apart Research × Formation Research Designed evaluation framework and measured its false-positive rate on a clean model: naive benchmark wrongly rejected 14/16 true nulls, but calibration-matched controls reduced this to 2/16. Top 25% of 179 submissions; one reviewer cited the calibration as “a key contribution”.  Apart Research project page ·  github.com/edimah/secret-loyalties Permutation Inference Multiple-Comparison Correction Pre-Registered Design	JULY 2026
ADVANCED METHODS FOR MODEL CALIBRATION WITH UNCERTAIN INPUTS M2 research thesis, EDF R&D, Chatou, solo Bayesian calibration of physical simulation models against noisy experimental data. Adaptive MCMC-based re-adjustment strategies, dubbed “MC × MCMC” and “quantisation × MCMC”; cut-distribution updates that respect the metrologists’ calibration chain; computational costs reported for each method. Bayesian Calibration MCMC Uncertainty Quantification Python OpenTURNS	APRIL – SEPTEMBER 2023
PROJECT REPORT – KERNEL METHODS FOR GRAPH DATA M2 MASH data challenge, group project Binary classification of molecules represented as graphs. Kernel methods implemented from scratch: n-th order walk kernel and kernel SVM. Kernel Methods SVM Graph Data Python	APRIL 2023
GIBBS MEASURES AND MCMC ALGORITHMS M1 research thesis, Paris Dauphine-PSL, group thesis supervised by Julien Stoehr Simulation of Gibbs measures with Markov chains: Metropolis-Hastings, Gibbs sampler, simulated annealing and Propp-Wilson algorithms. Theory, R implementation and oral defence. Markov Chains MCMC R LaTeX	2021 – 2022

EDUCATION

- 2023 **Master 2 – MASH** (Mathematics, Machine Learning, Sciences, Humanities) at **Université PSL** (Dauphine & ENS), with Merit (mention Bien).
Research thesis at EDF R&D: *Advanced methods for model calibration with uncertain inputs*.
- 2022 **Master 1 – Applied Mathematics**, Statistics track, at **Paris Dauphine-PSL**.
Group research thesis: *Mesure de Gibbs et algorithmes MCMC* (Gibbs Measures and MCMC Algorithms), supervised by Julien Stoehr.
- 2021 **Bachelor's – MEFA** (Mathematics, Economics, Finance, Actuarial Science) at **Paris Dauphine-PSL**
- 2025 *TOEFL iBT (112/120)*

HOBBIES

Interests Muay Thai · String instruments · Bouldering